

AGILE SOFTWARE ENGINEERING (CS303)

Waterfall vs Agile Comparison

Learning Objectives

Understand the Waterfall model, Agile methodology, and compare their characteristics, advantages, disadvantages, and applications.

Introduction to SDLC

Software Development Life Cycle (SDLC) provides a structured approach for software development. Waterfall and Agile are two widely used SDLC models.

Waterfall Model

Waterfall is a linear and sequential software development model where each phase must be completed before moving to the next phase.

Phases of Waterfall

Requirements → Design → Implementation → Testing → Deployment → Maintenance

Advantages of Waterfall

Simple to understand, easy documentation, suitable for stable requirements, and easier project management.

Disadvantages of Waterfall

Difficult to accommodate changes, testing occurs late, higher risk of failure if requirements change.

Agile Model

Agile follows an iterative and incremental approach with continuous feedback, customer collaboration, and frequent software releases.

Advantages of Agile

Faster delivery, flexibility, customer satisfaction, continuous testing, and reduced project risk.

Disadvantages of Agile

Requires active customer involvement, experienced teams, and continuous communication.

Waterfall vs Agile Comparison

1. Development Style: Sequential vs Iterative
2. Requirement Changes: Difficult vs Easy
3. Customer Involvement: Limited vs Continuous
4. Testing: End Phase vs Continuous
5. Delivery: Single Release vs Multiple Releases
6. Documentation: Extensive vs Lightweight
7. Risk Management: Higher vs Lower

8. Flexibility: Low vs High

Real World Examples

Waterfall is commonly used in construction and government projects. Agile is widely used by Google, Amazon, Spotify, and Netflix.

Exam Important Questions

1. Explain the Waterfall model.
2. Explain Agile methodology.
3. Compare Waterfall and Agile models.
4. Discuss advantages and disadvantages of Agile.
5. Which model is more suitable for dynamic requirements and why?

Summary

Waterfall is suitable for projects with stable requirements, while Agile is ideal for dynamic environments requiring flexibility and continuous customer feedback.